

5) Sea la Señal gaussiana: $x(t) = e^{-at^2}$, donde $a \in \mathbb{R}^+$:

$$\text{Sistema A: } y_A(t) = x^2(t)$$

$$\text{Sistema B: } h_B(t) = B e^{-bt^2}$$

Tomando en cuenta:

$$1) [X(\omega)] [h_B(\omega)] \longrightarrow Y(\omega)$$

$$2) y_A(t) \longrightarrow y'(t)$$

• Convolución de $(x(t))$ y $(h_B(t))$

$$y(t) = [x(t)] [h_B(t)] = \int_{-\infty}^{\infty} x(\tau) h_B(t-\tau) d\tau$$

Reemplazamos los valores en el sistema:

$$y(t) = \int_{-\infty}^{\infty} \underbrace{e^{a\tau^2}}_{x(\tau)} * \underbrace{B e^{-b(t-\tau)^2}}_{h_B(t-\tau)} d\tau$$

$$y(t) = B \int_{-\infty}^{\infty} e^{-a\tau^2} * e^{-b(t-\tau)^2} d\tau$$

$$y(t) = B \cdot \int_{-\infty}^{\infty} e^{-a\tau^2} \cdot e^{-b(t^2 - 2t\tau + \tau^2)} d\tau$$

$$y(t) = B \int_{-\infty}^{\infty} e^{-a\tau^2} e^{-b(t^2 - 2(-\tau + \tau^2))} d\tau$$

$$y(t) = B e^{-bt^2} \int_{-\infty}^{\infty} e^{-(a+b)\tau^2 + 2b\tau} d\tau$$

$$y(t) = B e^{-bt^2} \cdot \int_{-\infty}^{\infty} e^{-\frac{(a+b)}{2} \left[\tau^2 - \frac{2b}{a+b} \tau \right]} d\tau$$

$$\tau^2 - \frac{2bt}{a+b} \tau = \left(\tau - \frac{bt}{a+b} \right)^2 - \left(\frac{bt}{a+b} \right)^2$$

Substituímos:

$$y(t) = B e^{-bt^2} \int_{-\infty}^{\infty} e^{-(a+b)\tau} \left[\left(\tau - \frac{bt}{a+b} \right)^2 - \left(\frac{bt}{a+b} \right)^2 \right] d\tau$$

$$y(t) = B e^{-bt^2} e^{(a+b)\left(\frac{bt}{a+b}\right)^2} \int_{-\infty}^{\infty} e^{-(a+b)\left(\tau - \frac{bt}{a+b}\right)^2} d\tau$$

$$y(t) = B e^{-bt^2} e^{\frac{b^2 t^2}{a+b}} \int_{-\infty}^{\infty} e^{-(a+b)(\tau - \mu)^2} d\tau$$

Onde: $\mu = \frac{bt}{a+b}$

Si reemplazamos: $k = a+b$

La integral nos queda:

$$\Rightarrow \int_{-\infty}^{\infty} e^{-k(\tau - \mu)^2} d\tau = \sqrt{\frac{\pi}{k}}$$

Entonces:

$$y(t) = B \cdot e^{-bt^2 + \frac{b^2 t^2}{a+b}} \cdot \sqrt{\frac{\pi}{a+b}}$$

$$-bt^2 + \frac{b^2 t^2}{a+b} \Rightarrow -\frac{abt^2}{a+b}$$

Así:

$$y(t) = B \cdot e^{-\frac{abt^2}{a+b}} \cdot \sqrt{\frac{\pi}{a+b}}$$

Aplicamos: $Y_A(t) = Y^2(t)$

$$Y_A(t) = \left(B \cdot e^{-\frac{abt^2}{a+b}} \cdot \sqrt{\frac{\pi}{a+b}} \right)^2$$

$$\rightarrow Y_A(t) = B^2 e^{-2 \cdot \frac{abt^2}{a+b}} \cdot \frac{\pi}{a+b}$$

• Salida del S. s. t. Serie:

$$X(t) \longrightarrow Y_A(t) = X^2(t) \xrightarrow{h_B(t)} Y(t)$$

Aplicamos A dirichlet:

$$Y_A(t) = X^2(t) = (e^{-at^2})^2$$

$$Y_A(t) = e^{-2at^2}$$

Convolución con $h_B(t)$:

$$Y(t) = Y_A(t) * h_B(t)$$

$$Y(t) = \int_{-\infty}^{\infty} e^{-2a\tau^2} \cdot B e^{-b(t-\tau)^2} d\tau$$

$$Y(t) = B \int_{-\infty}^{\infty} e^{-2a\tau^2} \cdot e^{-b(b-\tau)^2} d\tau$$

$$\rightarrow Y(t) = B^2 e^{-2 \frac{abt^2}{2a+b}} \cdot \frac{\pi}{2a+b}$$

- Salida del sistema Serie:

$$X(t) \longrightarrow Y_A(t) = X^2(t) \longrightarrow Y(t)$$

Aplicamos A directamente:

$$Y_A = X^2(t) = (e^{-at})^2$$

$$Y_A = e^{-2at}$$

Convención con $h_B(t)$:

$$Y(t) = Y_A(t) * h_B(t)$$

$$Y(t) = \int_{-\infty}^{\infty} e^{-2a\tau} \cdot B e^{-b(t-\tau)^2} d\tau$$

$$Y(t) = B \cdot \int_{-\infty}^{\infty} e^{-2a\tau} \cdot e^{-b(b-\tau)^2} d\tau$$

$$\rightarrow Y(t) = B \cdot e^{-2 \frac{ab t^2}{2a+b}} \sqrt{\frac{\pi}{2a+b}}$$